AGRIVIVE / BAMBOOFLOW

MEGHALAYA / BAMBOO-DRIP INSPIRED / GRAVITY DISTRIBUTOR

SERIES 16 / 28

A3 / PRELIMINARY / ALL DIMENSIONS mm



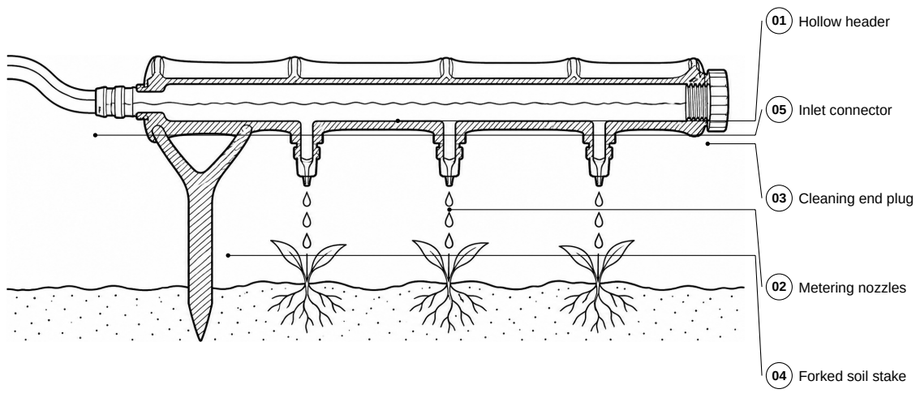
STATE CONTEXT / INDIGENOUS INSPIRATION

Bamboo drip networks in Meghalaya convey water by gravity through carefully arranged channels. BAMBOOFLOW proposes a small cleanable header with three replaceable outlets, inspired by that distribution principle.

Practice source: Meghalaya Basin Development Authority bamboo irrigation | Outline: geoBoundaries / schematic

01 INSTALLED SOIL / WATER SECTION

NOMINAL ENVELOPE / mm: 100 header L x 30 OD

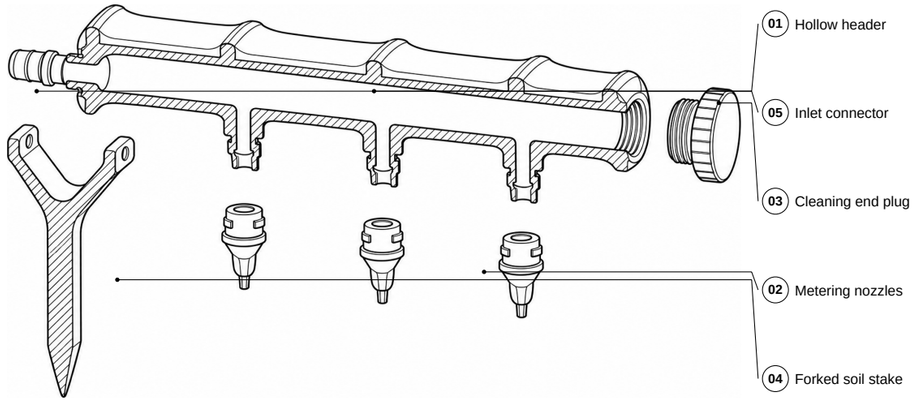


DESIGN / OPERATION NOTES

Support the horizontal header on the forked stake.
Connect an elevated gravity supply and fit the three nozzles.
Remove the end plug for flushing; actual drip rate requires tests.

02 EXPLODED ASSEMBLY

ASSEMBLED ENVELOPE / mm
100 header L x 30 OD

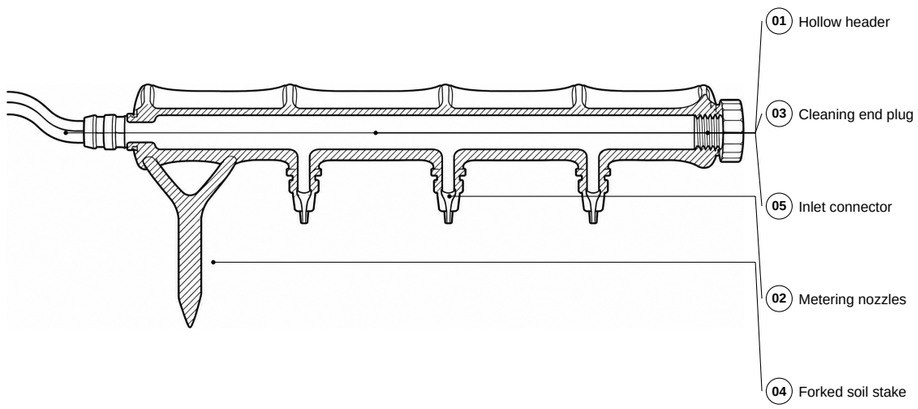


DESIGN / OPERATION NOTES

Components separated for service and assembly.
Connections and mating fits require CAD.

04 INTERNAL SECTION

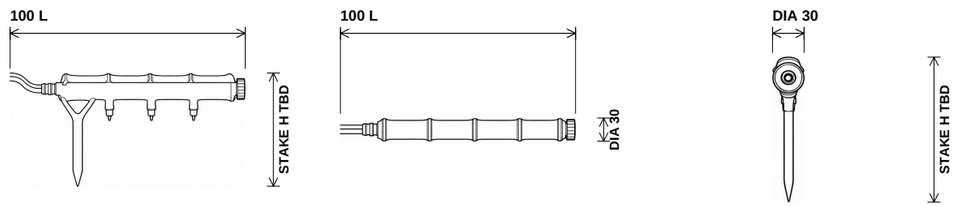
SECTION BASIS / mm
20 main bore ID; three 2 metering-port IDs



DESIGN / OPERATION NOTES

Connect an elevated gravity supply and fit the three nozzles.
Remove the end plug for flushing; actual drip rate requires tests.

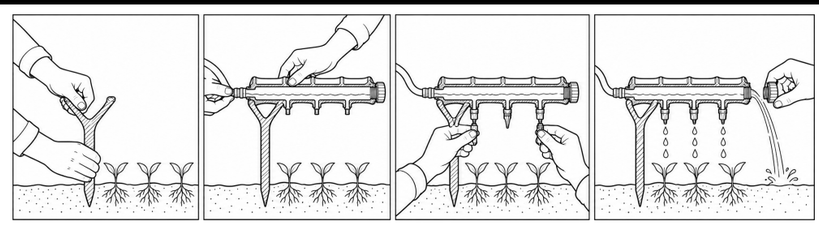
03 ORTHOGRAPHIC STUDIES / mm



FRONT / HEADER **TOP / HEADER** **SIDE / HEADER**

Header 100 L x DIA 30 OD / DIA 20 ID. Three DIA 2 nozzle bores; combined area does not establish drip rate.

05 INSTALLATION / OPERATING SEQUENCE



1 PLACE STAKE **2 FIT HEADER** **3 FIT NOZZLES** **4 FLUSH / CLEAN**

Prototype fit, access and operation require trials.

DESIGN / INTERFACE SCHEDULE

01 NOMINAL GEOMETRY

Header length / OD **100 / 30**

Main bore ID **20**

Three metering IDs **2 each**

02 GEOMETRIC CHECK

COMBINED METERING-PORT AREA

9.4 mm2

3 x pi/4 x 2^2
Drip rate depends on head, geometry and clogging

03 PARTS / QTY

01 Hollow header	**1**
02 Metering nozzles	**3**
03 Cleaning end plug	**1**
04 Forked soil stake	**1**
05 Inlet connector	**1**

Header/stake/nozzles: candidate polymer. Hose and connection material TBD.

04 INTERFACES / CHECKS

Test clogging, nozzle fit and flushing access.

Measure discharge at the intended supply head.

Check header support and hose retention.

05 DATUMS / DRAWING BASIS

A: header bore axis. B: nozzle seating faces; confirm each bore is continuous.
PRELIMINARY / VALIDATION PENDING

DESIGN DEVELOPMENT RECORD

01 / GEOMETRY
Nominal body sizes are proposed. Generated views are illustrative; small features and interfaces may vary.

02 / BEFORE FABRICATION
Develop one CAD model, assign wall thicknesses, tolerances and mating fits. Check print orientation and service access.

03 / TRIAL RECORD
Measure dimensions and retained volume or opening area. Record leakage, clogging and soil anchorage under observed conditions.

No discharge, treatment or durability performance is established.

DRAWING / REVISION	TITLE	STATUS	REVISION DATE	FORMAT / SCALE / UNITS	COLLECTION SHEET
AGR-16 / P01	BAMBOOFLOW	PRELIMINARY CONCEPT / CAD & TRIALS PENDING	10 NOV 2026	A3 / NTS / mm	16 / 28